

Variable-Horizon Model Predictive Control for Switched Systems

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Abstract

This paper investigates model predictive control (MPC) for switched systems subject to control and state constraints. A variable-horizon switched MPC approach is proposed. By steering the system state into a well-designed switching feasible set, the proposed method structurally decouples the dwell-time conditions from the MPC constraints, thereby relaxing the dwell-time requirements to match those of the unconstrained switched systems. Furthermore, algorithms are developed to construct this switching feasible set and characterize the domain of attraction, ensuring both persistent feasibility and closed-loop asymptotic stability. To further decouple the prediction horizon length from strict dwell-time bounds, advanced short-horizon switched MPC schemes are designed, which expand the overall domain of attraction. Simulations illustrate the efficacy of the proposed methods.

Key words: Switched system; Model predictive control; Variable-horizon, Domain of attraction; Dwell-time constraint

1 Introduction

A switched system consists of a family of subsystems and a logical rule that manages which subsystem is activated (Liberzon (2003)). In real-world applications, abrupt changes in structural parameters frequently occur because of intrinsic dynamics or external disturbances. Moreover, many systems operate in highly complex environments, exhibiting pronounced switching behaviors. Consequently, switched systems have emerged as a powerful modeling tool for a variety of applications such as bipedal robots (Liu et al. (2007)), aircraft engines (Shi and Sun (2021)), smart grids (Jaramillo-Cajica and Schiffer (2024)), and so on. It is well known that the stability of switched systems hinges on both the dynamics of the individual subsystems and the switching law. The switching signal (time-dependent, state-dependent, or hybrid) plays a critical role in determining whether the overall system is stable (Sun and Ge (2005)). Time-dependent switching strategies, such as dwell-time (Hespanha and Morse (1999)), average dwell-time (Hespanha and Morse (1999)), and mode-dependent average dwell-time (Zhao et al. (2011)), are widely used to guarantee

the system stability.

Another challenge in control design is the presence of physical constraints on control inputs and system states. Model predictive control (MPC) addresses this challenge by predicting the plant's behavior over a finite horizon and computing control actions that optimize a pre-defined performance index subject to these constraints (Shi and Zhang (2021)). Since MPC relies on anticipated state evolution, its implementation generally requires accurate system models and switching signal information.

Most early MPC schemes for switched systems assume that the switching signal is known *a priori* (Mhaskar et al. (2005)) or state-dependent switching (Abbasi et al. (2021)). Recently, infinite-horizon MPC laws have been co-designed with average dwell-time policies utilizing linear matrix inequality techniques (Nodozi and Rahmani (2017)). Because the prediction horizon is inherently infinite, these approaches implicitly circumvent transient feasibility issues by leveraging global Lyapunov-theoretic invariant sets. Stabilizing MPC without terminal constraints has also been investigated (Knüfer et al. (2016)). Recent advances in this controlled switching paradigm seek to jointly optimize the control input and the switching signals under dwell-time constraints (Chen and Lazar (2022); Zhuang et al. (2023)). Although effective, these active switching strategies typically require solving computationally demanding mixed-integer optimization problems.

In contrast, for strictly exogenous switching scenarios where the switching signal is unknown in advance and dictated by external environments, the controller cannot co-optimize the switching logic but must survive it.

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Guaranteeing the persistent feasibility of finite-horizon MPC under exogenous switching is notoriously challenging, as the pre- and post-switching models may differ drastically. For instance, while Bridgeman et al. (2016) developed a variable terminal set framework to ensure recursive feasibility, it strictly requires the switching sequence to be known in advance. When the switching is unknown, a key requirement is that the state at each switching instant must successfully transition into the feasible region of the subsequent mode. As demonstrated by Zhang and Braatz (2013), this creates a circular dependency: the dwell-time requirement depends on the switching-instant state, yet the latter must itself satisfy the constraints dictated by the dwell-time condition.

To break this circular dependency, the foundational work of Zhang et al. (2016) established rigorous links among feasibility, attractivity, and stability via backward reachable set analysis. By enforcing rigid multi-step geometric intersections, the system state is compelled into the one-step reachable set at switching instants. While recent works of Tan et al. (2024) have extended this framework to broader contexts, the underlying geometric mechanism remains identical. The price of this fixed-horizon geometric approach is twofold: (i) the computational complexity grows with the system dimension and the number of mode, and (ii) the resulting dwell-time constraints are excessively restrictive.

Taking a different approach, a min-max switching MPC scheme developed by Ong et al. (2016) exploits a truncated admissible switching sequence. However, this min-max paradigm suffers from severe online computational demands and inherently yields a significantly restricted domain of attraction. Alternatively, other recent strategies ensure recursive feasibility by constructing switch-robust control invariant (switch-RCI) sets (Danielson et al. (2019)). Lavaei et al. (2020) further extended this idea to handle bounded disturbances under a tube-based framework. However, these studies focus on constrained enforcement rather than asymptotic convergence.

Consequently, existing exogenous switching frameworks are highly difficult to extend to higher-dimensional or large-scale systems, owing to overly conservative offline dwell-time conditions, prohibitive online computational complexity, or the lack of rigorous stability guarantees. To address the above-discussed challenging issues, this paper proposes a *variable-horizon* switched MPC strategy for switched systems subject to mode-dependent dwell-time constraints. The contributions of this paper are three-fold:

i) A novel *variable-horizon* switched MPC framework is proposed, which structurally decouples the dwell-time conditions from the stringent MPC state and input constraints. By actively steering the system state into a predefined switching feasible set, this framework aligns the dwell-time constraints with those required for unconstrained switched systems. Compared to the method of Zhang et al. (2016), which enforces strict dwell-time constraints to force persistent feasibility, our technique

significantly reduces conservatism and provides structurally simpler conditions.

ii) Effective algorithms are developed to construct the switching feasible set and explicitly characterize the domain of attraction. Empowered by the design of the *variable-horizon* mechanism and the switching feasible set, the proposed method guarantees persistent feasibility at switching instants alongside closed-loop asymptotic stability. Unlike the exhaustive min-max paradigm of Ong et al. (2016), our approach requires a shorter prediction horizon and eliminates additional input restrictions, yielding a larger feasible region. Furthermore, it overcomes the theoretical limitations of Danielson et al. (2019) by providing asymptotic stability guarantees.

iii) Short-horizon switched MPC schemes are used to decouple the required prediction horizon length from dwell-time. By tailoring the dwell-time constraint specifically for the initial mode transition, these schemes circumvent the geometric limitations of short horizons. Enlarging initialization dwell time expands the domain of attraction without sacrificing the relaxed dwell-time benefits for subsequent transitions.

Notation: For a matrix A , $A^\top$ denotes its transpose. The sets of real numbers and positive integers are denoted by $\mathbb{R}$ and $\mathbb{Z}_+$ respectively. The discrete interval of integers from a to b is represented by $\mathbb{Z}_{[a,b]}$. For a vector x and positive definite matrix P , the weighted Euclidean norm is defined as $\|x\|_P \triangleq \sqrt{x^\top P x}$. Symbols $\succ$ and $\succeq$ imply positive definite and positive semi-definite. A sequence of variables from index a to b is denoted by $\{x_i\}_{i=a}^b \triangleq \{x_a, x_{a+1}, \dots, x_b\}$.

The remainder of this paper is organized as follows. Section 2 gives some preliminaries and formulates the variable-horizon switched MPC problem. Section 3 develops the algorithm for the switching feasible set and establishes the persistent feasibility of the proposed approach. Section 4 computes the domain of attraction and proves the asymptotic stability of the system under the derived mode-dependent dwell-time constraints. To further mitigate the restrictive impact of prediction horizons on dwell-time constraints, Section 5 proposes short-horizon switched MPC schemes. Numerical simulations are presented in Section 6 to support the obtained results. Finally, we conclude this paper in Section 7. To facilitate understanding, the overall framework and the logical relationships among these sections are illustrated by Fig. 1.

2 Problem Formulation

Consider a class of discrete-time switched linear systems:

$$(\Omega_{\sigma(k)}) : x_{k+1} = A_{\sigma(k)}x_k + B_{\sigma(k)}u_k, \quad (1)$$

where $x_k \in \mathbb{R}^{n_x}$ denotes the system state and $u_k \in \mathbb{R}^{n_u}$ represents the control input. $\sigma(k) \in \mathcal{M} = \{1, \dots, M\}$ is the switching signal with $M > 1$ being the number of subsystems. Here the switching signal $\sigma(k)$ is unknown in advance. Assume that (A_m, B_m) is stabilizable

Core results

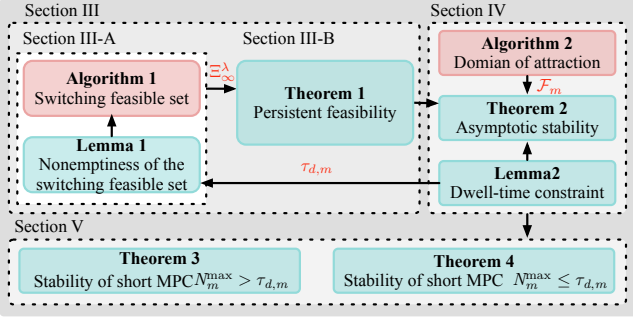


Fig. 1. Schematic illustrating the relation of the presented algorithms and theories.

for every m . The state and input are subject to mode-dependent constraints: $x_k \in \mathbb{X}_m$ and $u_k \in \mathbb{U}_m$, where $\mathbb{X}_m \subseteq \mathbb{R}^{n_x}$ and $\mathbb{U}_m \subseteq \mathbb{R}^{n_u}$ are compact polyhedral sets containing the origin in their interiors.

Definition 1 (Dehghan and Ong (2013)) For any two consecutive switching instants k_s and k_{s+1} with $s \in \mathbb{N}_+$, if $k_{s+1} - k_s \geq \tau_{d,m} > 0$ where $m = \sigma(k_s^+) \in \mathcal{M}$, then $\tau_{d,m}$ is called the mode-dependent dwell-time for the m -th subsystem.

Let $\mathcal{D}(k) = k - k_s$ be the duration of the current mode $m = \sigma(k)$, where k_s is the latest switching instant. The variable prediction horizon is defined as

$$N_m(k, \mathcal{D}(k)) = \max \{N_m^{\max} - \mathcal{D}(k), N_m^{\min}\}, \quad (2)$$

where $N_m^{\min}$ and $N_m^{\max}$ are the minimum and maximum horizons, respectively. For brevity, let $N_k^m \triangleq N_m(k, \mathcal{D}(k))$. Define $\Delta N_m \triangleq N_m^{\max} - N_m^{\min}$ as the horizon range.

Cost function for $m \in \mathcal{M}$ is $J_m(x_k, \mathbf{u}_k) \triangleq T_m(x_{N_k^m|k}) +$

$\sum_{i=0}^{N_k^m-1} \ell_m(x_{i|k}, u_{i|k})$. $\mathbf{u}_k \triangleq \{u_{0|k}, \dots, u_{N_k^m-1|k}\}$ denotes the predicted control sequence. $u_{i|k}$ is the predicted control input at instant $k+i$. The symbol $x_{i|k}$ for $i \in \{1, \dots, N_k^m\}$ represents the state at instant $k+i$ under the control sequence $\mathbf{u}_k$. The cost functions are $\ell_m(x_{i|k}, u_{i|k}) = \|x_{i|k}\|_{Q_m}^2 + \|u_{i|k}\|_{R_m}^2$, $T_m(x_{N_k^m|k}) = \|x_{N_k^m|k}\|_{P_m}^2$, where $Q_m \in \mathbb{R}^{n_x \times n_x} \succ 0$, $R_m \in \mathbb{R}^{n_u \times n_u} \succ 0$ and $P_m \in \mathbb{R}^{n_x \times n_x} \succ 0$ are state weighting, input weighting and terminal state weighting matrix, respectively. Here, to ensure closed-loop stability, the terminal weighting matrix P_m is chosen as the positive definite matrix solution to the DARE: $P_m = A_m^\top P_m A_m - A_m^\top P_m B_m (R_m + B_m^\top P_m B_m)^{-1} B_m^\top P_m A_m + Q_m$. Let K_m be the controller gain for the m -th subsystem in the form of $K_m = -(R_m + B_m^\top P_m B_m)^{-1} B_m^\top P_m A_m$. Additionally, let $\bar{A}_m \triangleq A_m + B_m K_m$.

The model predictive control optimization problem is formulated as follows:

$$\mathbf{u}_k^* = \arg \min_{\mathbf{u}_k} J_m(x_k, \mathbf{u}_k), \quad (3a)$$

$$\text{s.t. } x_{i+1|k} = A_m x_{i|k} + B_m u_{i|k}, x_{0|k} = x_k, \quad (3b)$$

$$x_{i|k} \in \mathbb{X}_m, u_{i|k} \in \mathbb{U}_m, \forall i \in \mathbb{Z}[0, N_k-1], \quad (3c)$$

$$x_{i|k} \in \Theta(\lambda, s), \forall i \in \mathbb{Z}[N_k - N_m^{\min}, N_k-1], \quad (3d)$$

$$x_{N_k|k} \in \mathcal{T}_\infty, \quad (3e)$$

where $\mathbf{u}_k^*$ stands for the optimal control sequence for the current mode $m = \sigma(k)$. $\Theta(\lambda, s)$ denotes the contracted switching feasible set parameterized by λ with s indicating the number of switches. $\mathcal{T}_\infty$ represents the terminal constraint set presented in Dehghan and Ong (2012). Due to space limitations, this part is omitted here.

Fig. 2 illustrates the multi-mode prediction paradigm for a second-order system at a switching instant k_s and

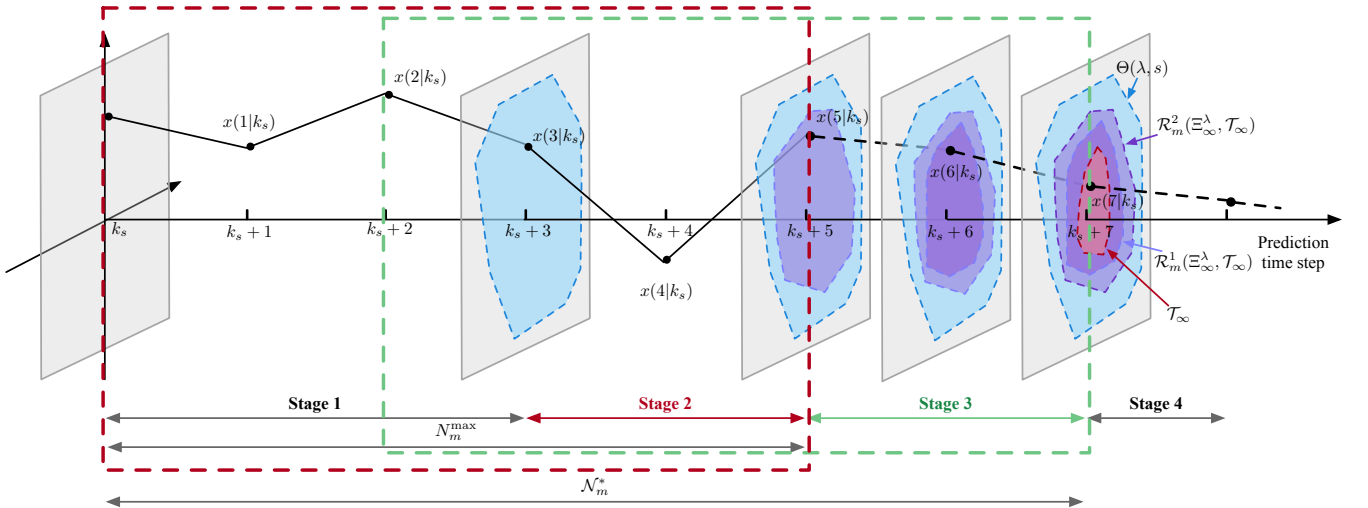


Fig. 2. Prediction paradigm of the variable-horizon switched MPC at a switching instant k_s (left) and $k_s + 3$ (right) with $N_m^{\max} = 5$ and $N_m^{\min} = 2$. At k_s , the control sequence steers the state into the contracted switching feasible set $\Theta(\lambda, s)$ within $N_m^{\max} - N_m^{\min}$ steps, maintains it therein, and finally drives it into the terminal set $\mathcal{T}_\infty$. At $k_s + 3$, the horizon reaches its minimum, reducing to a fixed-horizon MPC.

Table 1

Prediction horizon and state constraints at each instant and predicted step with $N_m^{\max} = 5$ and $N_m^{\min} = 2$ (see (3))

	k_s	$k_s + 1$	$k_s + 2$	$[k_s + 3, k_{s+1})$
Horizon	5	4	3	2
Step 1	$\mathbb{X}_m$	$\mathbb{X}_m$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$
Step 2	$\mathbb{X}_m$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\mathcal{T}_\infty$
Step 3	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\mathcal{T}_\infty$	N/A
Step 4	$\Theta(\lambda, s)$	$\mathcal{T}_\infty$	N/A	N/A
Step 5	$\mathcal{T}_\infty$	N/A	N/A	N/A

three steps later at $k_s + 3$ (equivalent to a fixed-horizon MPC). The gray, blue, and red areas represent the state constraint $\mathbb{X}_{\sigma(k_s)}$, the contracted switching feasible set $\Theta(\lambda, s)$, and the terminal set $\mathcal{T}_\infty$, respectively. Resetting the horizon to $N_m^{\max}$ at each switch guarantees that the state enters the contracted switching feasible set $\Theta(\lambda, s)$ within $N_m^{\max} - N_m^{\min}$ steps and reaches $\mathcal{T}_\infty$ within $N_m^{\max}$ steps. The detailed step-by-step evolution of the variable horizon and the corresponding state constraints are summarized in Table 1.

When $\mathcal{D}(k) = \Delta N_m$, the proposed MPC problem (3) reduces to a fixed-horizon MPC strategy

$$\mathbf{u}_k^* = \arg \min_{\mathbf{u}_k} J_m(x_k, \mathbf{u}_k), \quad (4a)$$

$$\text{s.t. } x_{i+1|k} = A_m x_{i|k} + B_m u_{i|k}, \quad x_{0|k} = x_k, \quad (4b)$$

$$x_{i|k} \in \Theta(\lambda, s), \quad u_{i|k} \in \mathbb{U}_m, \quad \forall i \in \mathbb{Z}_{[0, N_m^{\min}-1]}, \quad (4c)$$

$$x_{N_m^{\min}|k} \in \mathcal{T}_\infty. \quad (4d)$$

Here, the state constraints are simplified to $\Theta(\lambda, s)$, and the terminal state strictly enters $\mathcal{T}_\infty$ at step $N_m^{\min}$. As illustrated by Stage 2 in Fig. 2, constraint (3d) represents the core mechanism of the variable-horizon scheme. It is exactly this mechanism that guarantees persistent feasibility at future switching instants as will be rigorously analyzed in Section 3.

The primary objectives of this paper are threefold: 1) To design the contracted switching feasible set $\Theta(\lambda, s)$; 2) To ensure persistent feasibility; 3) To compute the domain of attraction that guarantees stability for the variable-horizon switched MPC problem (3).

3 Persistent Feasibility

Persistent feasibility is a fundamental challenge in the MPC problem. For non-switched systems, it is guaranteed by driving the terminal state into a positive invariant set. However, unknown future switching signals render this standard approach invalid for switched systems, where feasibility is often vulnerable at the exact switching instants. Motivated by the capability of variable-horizon MPC to steer states into a target set in finite time (Richards and How (2006)), this section rigorously addresses this challenge. By incorporating dwell-time constraints, a dedicated *contracted switching feasible set* $\Theta(\lambda, s)$ is constructed, which effectively reformulates the feasibility requirement at switching instants and guarantees persistent feasibility for the proposed MPC problem (3).

3.1 Switching Feasible Set

Let $\mathcal{P}_m(\mathcal{S}) \triangleq \{x \in \mathbb{X}_m \mid \exists u \in \mathbb{U}_m, A_m x + B_m u \in \mathcal{S}\}$ denote the one-step backward reachable set (pre-set) for subsystem m . Its l -step recursive application is denoted by $\mathcal{P}_m^l(\mathcal{S})$, with $\mathcal{P}_m^0(\mathcal{S}) \triangleq \mathcal{S}$. Define $\mathbb{T}_0^m \triangleq \{0, 1, \dots, \tau_{d,m} - 1\}$.

Algorithm 1 Computation of switching feasible set

Input: $\lambda, A_m, B_m, \mathbb{X}_m, \mathbb{U}_m, \mathcal{T}_\infty, N_m^{\max}$ and $N_m^{\min}$
Output: Ξ_∞^λ

- 1: Set $k = 0$ and let $\Xi_0^\lambda = \cap_{m \in \mathcal{M}} \mathbb{X}_m \cap_{t \in \mathbb{T}_0^m} \mathcal{P}_m^t(\mathbb{X}_m)$
- 2: **while** $\Xi_{k+1}^\lambda \neq \Xi_k^\lambda$ **do**
- 3: $\Xi_{k+1}^\lambda = \Xi_k^\lambda$
- 4: **for** $m \in \mathcal{M}$ **do**
- 5: $\mathcal{W}_0 = \mathcal{T}_\infty$
- 6: **for** $t = 1$ to $N_m^{\min}$ **do**
- 7: $\mathcal{W}_t = \mathcal{P}_m^1(\mathcal{W}_{t-1}) \cap \lambda \Xi_k^\lambda$
- 8: **end for**
- 9: $\Xi_{k+1}^\lambda = \Xi_{k+1}^\lambda \cap \mathcal{P}_m^{N_m^{\max}-N_m^{\min}}(\mathcal{W}_{N_m^{\min}})$
- 10: **end for**
- 11: $k = k + 1$
- 12: **end while**
- 13: $\Xi_\infty^\lambda = \Xi_{k+1}^\lambda$

Algorithm 1 systematically constructs the switching feasible set, thereby formally guaranteeing the persistent feasibility of the proposed MPC scheme. Specifically, Step 1 initializes the set to enforce state constraints up to the dwell-time boundary. The iterative backward reachability in Steps 4–9 rigorously ensures that for any $x_k \in \Xi_\infty^\lambda$, an admissible control sequence exists such that the predicted trajectory: (i) satisfies the common state constraints $\cap_{m \in \mathcal{M}} \mathbb{X}_m$ during the initial steps $i \in [1, N_m^{\max} - N_m^{\min} - 1]$; (ii) remains strictly within the contracted feasible set $\lambda \Xi_\infty^\lambda$ during the interval $i \in [N_m^{\max} - N_m^{\min}, N_m^{\max} - 1]$; and (iii) reaches the terminal set $\mathcal{T}_\infty$ exactly at step $N_m^{\max}$.

Proposition 1 *The set Ξ_∞^λ generated by Algorithm 1 satisfies: (i) $\Xi_\infty^\lambda \subseteq \mathbb{X}_m$ and $\Xi_{j+1}^\lambda \subseteq \Xi_j^\lambda, \forall j \geq 1$; (ii) Ξ_∞^λ is compact and contains the origin; (iii) $\Xi_\infty^{\bar{\lambda}} \subseteq \Xi_\infty^\lambda$ for $\bar{\lambda} \leq \lambda \leq 1$; and (iv) $\mathcal{T}_\infty \subseteq \Xi_\infty^\lambda$.*

The proof follows directly from the iterative construction in Algorithm 1 and is thus omitted.

Lemma 1 *If the unconstrained switched system (1) using controller gain K_m is asymptotically stable under the mode-dependent dwell-time constraints $\{\tau_{d,m}\}$, then there exists a constant $\lambda \in (0, 1)$ such that both the switching terminal set $\mathcal{T}_\infty$ and the switching feasible set Ξ_∞^λ are non-empty.*

PROOF. By (Dehghan and Ong, 2012, Theorem 6), the stated asymptotic stability guarantees a non-empty set $\mathcal{T}_\infty$ for some $\lambda \in (0, 1)$. It then follows from Proposition 1 (iv) that $\mathcal{T}_\infty \subseteq \Xi_\infty^\lambda$, which directly ensures $\Xi_\infty^\lambda \neq \emptyset$. $\square$

Remark 1 While Ξ_∞^λ is a subset of $\bigcap_{m \in \mathcal{M}} \mathcal{P}_m^{N_m^{\max}}(\mathcal{T}_\infty)$, there is no explicit inclusion relationship between the contracted set $\Theta(\lambda, s)$ and the standard $N_m^{\min}$ -step domain of attraction $\bigcap_{m \in \mathcal{M}} \mathcal{P}_m^{N_m^{\min}}(\mathcal{T}_\infty)$. A detailed geometric analysis and visual demonstration of this property will be provided in Section 6.

Remark 2 The switching feasible set Ξ_∞^λ differ from the exact switch-RCI framework of Danielson et al. (2019). In Danielson et al. (2019), each mode-dependent switch-RCI set is required to be one-step control invariant under the corresponding mode; thus, the state must remain in that set once it enters. This stepwise containment severely restricts the allowable state space during offline computation. In contrast, our approach explicitly couples the set construction with the variable-horizon MPC mechanism. By resetting the prediction horizon at switching instants, the MPC provides a transient buffer of ΔN_m steps. This co-design completely bypasses the conservative one-step invariance requirement, replacing it with a multi-step backward reachability condition that explicitly exploits the mode-dependent dwell time. Consequently, intermediate states may temporarily evolve outside Ξ_∞^λ (provided that they remain within the physical bounds of $\mathbb{X}_m$). Furthermore, rather than merely guaranteeing the hard constraints in standard RCI frameworks, our scheme guarantees closed-loop asymptotic stability, as the terminal constraint ultimately drives the state into the terminal set $\mathcal{T}_\infty$.

3.2 Persistent Feasibility

This subsection proves the persistent feasibility of the proposed variable-horizon switched MPC (3).

Theorem 1 Suppose the MPC problem (3) is feasible at the initial instant, $\Delta N_m \leq \tau_{d,m}$ for all $m \in \mathcal{M}$, and $\Theta(\lambda, s) = \lambda^s \Xi_\infty^\lambda$ where $s \in \mathbb{Z}_{[0, \infty]}$ is the switch count. Then, (3) is persistently feasible for all future instants.

PROOF. Let $\mathbf{x}_{k_0}^* = \{x_{i|k_0}^*\}_{i=1}^{N_{k_0}^m}$ and $\mathbf{u}_{k_0}^* = \{u_{i|k_0}^*\}_{i=0}^{N_{k_0}^m-1}$ be the optimal sequences at k_0 for mode $m = \sigma(k_0)$. At instant $k_0 + 1$, one has $x_{0|k_0+1} = x_{1|k_0}^*$ and $N_{k_0+1}^m = N_{k_0}^m - 1$. Hence, the sequences $\hat{\mathbf{x}}_{k_0+1} = \{x_{i|k_0+1}^*\}_{i=2}^{N_{k_0}^m}$ and $\hat{\mathbf{u}}_{k_0+1} = \{u_{i|k_0+1}^*\}_{i=1}^{N_{k_0}^m-1}$ remain feasible. Iterating this process maintains feasibility over $[k_0, k_0 + \Delta N_m]$. For $k \in [k_0 + \Delta N_m, k_1]$, (3) reduces to the standard fixed-horizon MPC (4). Since the terminal state enters the invariant set $\mathcal{T}_\infty$, recursive feasibility is inherently guaranteed by appending the local controller K_m . At the first switch k_1 , we have $x_{k_1} \in \Theta(\lambda, 0)$. Algorithm 1 guarantees a feasible control sequence exists at this instant. By induction, the problem remains persistently feasible for all future intervals and switching instants. $\square$

Remark 3 Theorem 1 establishes a direct link between the variable horizon and dwell-time constraints. Since future switching signals are unknown, traditional methods like Zhang et al. (2016) must guarantee feasibility under worst-case scenarios, relying on computationally

intensive set operations to derive minimum dwell-times. Our approach reverses this paradigm: it utilizes predefined dwell-times to proactively design the horizon length. By actively steering states into predefined switching feasible set within the prescribed dwell time, the variable-horizon mechanism bypasses these complex recursive calculations. This yields a twofold advantage: a significant reduction in computational complexity and a substantial improvement in real-world deployability.

4 Stability

By Theorem 1, persistent feasibility hinges on a non-empty switching feasible set Ξ_∞^λ and adequate dwell-times. This section first derives the dwell-time conditions ensuring $\Xi_\infty^\lambda \neq \emptyset$. We then characterize the domain of attraction and establish asymptotic stability.

Since K_m stabilizes each subsystem, there exist $\rho_m > 0$ and $\lambda_m \in (0, 1)$ such that $\|(A_m + B_m K_m)^n\| \leq \rho_m \lambda_m^n$ for all $m \in \mathcal{M}$.

Lemma 2 The unconstrained switched system (1) under controllers $u_k = K_{\sigma(k)} x_k$ is asymptotically stable if the dwell-times $\tau_{d,m}$ satisfy

$$\rho_m \lambda_m^{\tau_{d,m}} < 1, \quad \forall m \in \mathcal{M}. \quad (5)$$

PROOF. For any consecutive switches k_s and k_{s+1} with mode $m = \sigma(k_s)$, the state evolves as $x_{k_{s+1}} = (A_m + B_m K_m)^{k_{s+1} - k_s} x_{k_s}$. Taking the norm and applying $k_{s+1} - k_s \geq \tau_{d,m}$ yields $\|x_{k_{s+1}}\| \leq \rho_m \lambda_m^{\tau_{d,m}} \|x_{k_s}\|$. Condition (5) ensures a strict contraction $\|x_{k_{s+1}}\| < \|x_{k_s}\|$ at each switch. By recursion, $\lim_{k \rightarrow \infty} \|x_k\| = 0$, establishing asymptotic stability. $\square$

Algorithm 2 Computation of the Domain of Attraction

Input: $A_m, B_m, \mathbb{X}_m, \mathbb{U}_m, \Xi_\infty^\lambda, \mathcal{T}_\infty, m, N_m^{\max}, N_m^{\min}$
Output: $\mathcal{F}_m$
1: $\mathcal{W}_0^m = \mathcal{T}_\infty$
2: **for** $t = 1$ to $N_m^{\min}$ **do**
3: $\mathcal{W}_t^m = \mathcal{W}_{t-1}^m \cup (\mathcal{P}_m^1(\mathcal{W}_{t-1}^m) \cap \Xi_\infty^\lambda)$
4: **end for**
5: $\mathcal{F}_m = \mathcal{W}_{N_m^{\min}}^m \cup \mathcal{P}_m^{N_m^{\max} - N_m^{\min}}(\mathcal{W}_{N_m^{\min}}^m)$

Algorithm 2 outlines the computation of the domain of attraction $\mathcal{F}_m$ for (3). Problem (3) is initially feasible for mode m if $x_0 \in \mathcal{F}_m$. Notably, the set Ξ_∞^λ constitutes a subset of each mode-specific feasible set, i.e., $\Xi_\infty^\lambda \subseteq \bigcap_{m \in \mathcal{M}} \mathcal{F}_m$.

Theorem 2 Consider the MPC problem (3) for the switched system (1) with horizon satisfying (2). Suppose $\Delta N_m \leq \tau_{d,m}$ for all $m \in \mathcal{M}$, the dwell-times $\tau_{d,m}$ satisfy (5), and the initial state $x_0 \in \mathcal{F}_{\sigma(0)}$ is computed via Algorithm 2. Then, the closed-loop switched system under (3) is asymptotically stable.

PROOF. Feasibility: Lemmas 1 and 2 guarantee that $\mathcal{T}_\infty$ and Ξ_∞^λ are nonempty for some $\lambda \in (0, 1)$. Since $x_0 \in \mathcal{F}_{\sigma(0)}$, problem (3) is initially feasible, and its persistent feasibility follows from Theorem 1.

Stability: Let $V_k = J_{\sigma(k)}^*(x_k)$ denote the optimal value of problem (3). Consider two consecutive instants k and $k+1$ within the same mode m . During the decreasing-horizon phase, removing the first element of the optimal sequence at k gives a feasible candidate at $k+1$. Once the horizon reaches $N_m^{\min}$, a feasible candidate is obtained by shifting the optimal sequence and appending the terminal control K_mx . From the DARE identity, $T_m((A_m + B_m K_m)x) - T_m(x) = -\ell_m(x, K_mx)$. Consequently, $V_{k+1} - V_k \leq -\ell_m(x_k, u_k^*)$.

Let k_s denote the switching instants. By constraint (3d), Algorithm 1, and $\Delta N_m \leq \tau_{d,m}$, $x_{k_{s+1}} \in \Theta(\lambda, s) = \lambda^s \Xi_\infty^\lambda$. Because the dynamics are linear, the cost is quadratic, and the feasible transition constructed by Algorithm 1 can be scaled by λ^s , there exists a constant $c_V > 0$, independent of s , such that $V_{k_{s+1}} \leq c_V \lambda^{2s}$.

If infinitely many switches occur, then for every $k \in [k_{s+1}, k_{s+2})$, we have $V_k \leq c_V \lambda^{2s}$. Moreover, with $\underline{q} = \min_{m \in \mathcal{M}} \lambda_{\min}(Q_m) > 0$, one has $\underline{q} \|x_k\|^2 \leq V_k \leq c_V \lambda^{2s}$. Since $\lambda \in (0, 1)$, this implies $x_k \rightarrow 0$.

If only finitely many switches occur, the system eventually remains in one fixed mode. After the final switch, one has $\sum_{k=k_s}^\infty \ell_m(x_k, u_k^*) \leq V_{k_s} < \infty$. Since $Q_m \succ 0$, it follows that $x_k \rightarrow 0$. Therefore, the closed-loop system is asymptotically stable. $\square$

Remark 4 Compared with Zhang et al. (2016), the proposed dwell-time condition has a simpler subsystem-wise form and yields less conservative bounds. Through the co-design of the variable prediction horizon and the switching feasible set, feasibility under the hard state and input constraints is handled by the horizon and set conditions. Once the stabilizing feedback gain K_m is obtained from the DARE for the chosen LQR weighting matrices, the transient factor ρ_m and decay rate λ_m are determined from $|(A_m + B_m K_m)^n| \leq \rho_m \lambda_m^n$. Condition of $\rho_m \lambda_m^{\tau_{d,m}} < 1$ remains sufficient whose conservatism depends on the selected feedback gain, norm, and exponential estimate, but it is not further restricted by explicit hard-constraint or cross-mode set calculations.

Remark 5 The proposed framework can be extended to a robust tube-based MPC formulation for switched systems subject to bounded additive disturbances. Following Mayne et al. (2005), for each mode m , a local gain K_m and a robust positively invariant set $\mathbb{Z}_m$ can be constructed for the error dynamics, yielding the tightened nominal constraints of $\hat{\mathbb{X}}_m = \mathbb{X}_m \ominus \mathbb{Z}_m$ and $\hat{\mathbb{U}}_m = \mathbb{U}_m \ominus K_m \mathbb{Z}_m$. The predecessor-set computations by Algorithms 1 and 2 are then performed for the nominal system using these tightened constraint sets to construct the corresponding nominal terminal and switching feasible sets. Let $\hat{\Theta}_m(\lambda, s)$ denote a robust nominal switching target set to be designed. Its construction should jointly address constraint tightening, inter-step reachability or invariance, tube re-centering under admissible mode transitions, and an appropriate practical stability condition. Since the mode-dependent tube does not contract with λ ,

nominal-set scaling alone could not establish contraction of the actual-state set or closed-loop stability. The detailed set construction and practical-stability analysis are left for future investigation.

Remark 6 The variable-horizon mechanism introduces a trade-off between closed-loop performance and switching feasibility. Shortening the prediction horizon reduces the prediction capability and may therefore degrade performance. However, this effect can be mitigated because the full prediction horizon is used immediately after each switch and shortens only as the state is driven toward the switching feasible set. The DARE-based terminal weight P_m further mitigates the truncation effect by accounting for the cost beyond the shortened horizon; it represents the exact infinite-horizon tail cost for the corresponding unconstrained fixed-mode LQR problem.

The horizon parameters distinctly shape the switching feasible set $\Xi_\infty^\lambda \subseteq \bigcap_{m \in \mathcal{M}} \mathcal{P}_m^{N_m^{\max}}(\mathcal{T}_\infty)$. Increasing $N_m^{\max}$ while fixing $N_m^{\min}$ enlarges the span ΔN_m , thereby expanding the feasible set. Conversely, increasing $N_m^{\min}$ with a fixed $N_m^{\max}$ narrows ΔN_m , creating a trade-off where stricter transition conditions may paradoxically shrink the domain of attraction. Furthermore, synchronously shifting the horizon window, i.e., increasing both $N_m^{\max}$ and $N_m^{\min}$ while keeping ΔN_m constant, initially expands the switching feasible set. However, this growth saturates beyond a critical threshold, as the feasible space is ultimately bounded by the system's inherent maximal stabilizable set. More importantly, blindly pursuing a larger domain of attraction by increasing $N_m^{\max}$ inevitably exacerbates the computational burden and demands longer mode-dependent dwell-time (since $\Delta N_m \leq \tau_{d,m}$). To break this geometric and computational bottleneck, the following section introduces a short-horizon switched MPC scheme, designed to effectively enlarge the feasible region without relying on excessively long prediction horizons.

5 Short-Horizon Switched MPC

The feasible set is determined by the horizon bounds $N_m^{\max}$, $N_m^{\min}$, and the switching feasible set. Fixing the prediction horizon at $N_m^{\max}$ as in Zhang et al. (2016) shrinks Ξ_∞^λ . Consequently, the overall feasible set of the proposed variable horizon MPC is smaller than that in Zhang et al. (2016). In addition, the variable horizon MPC requires $N_m^{\max} \geq \tau_{d,m}$ and an upper bound of $N_m^{\max} \leq \tau_{d,m} + N_m^{\min}$. If the predefined dwell time is close to the maximum horizon limit, the required minimum horizon $N_m^{\min}$ cannot be sufficiently small, which severely restricts the feasible set. To mitigate this restriction, we propose a short horizon switched MPC scheme for two cases: (i) $N_m^{\max} \geq \tau_{d,m}$ and (ii) $N_m^{\max} < \tau_{d,m}$.

Let N_m^* and $N_m^{\max}$ denote the desired and actual maximum prediction horizons, respectively. Define N_m^* as the desired minimum prediction horizon within the switching feasible set. The relationship between N_m^* and $N_m^{\min}$ is analogous to that between N_m^* and $N_m^{\max}$.

The state enters the terminal set $\mathcal{T}_\infty$ within $\mathcal{N}_m^*$ steps, and it reaches and stays in the switching feasible set in exactly $\underline{\mathcal{N}}_m^*$ steps after a switch. For set computations, define a generalized predecessor operator $\mathcal{R}_m(\mathbb{A}, \mathcal{S}) \triangleq \{x \in \mathbb{A} \mid \exists u \in \mathbb{U}_m, A_m x + B_m u \in \mathcal{S}\}$. Its recursive application is denoted by $\mathcal{R}_m^l(\mathbb{A}, \mathcal{S}) = \mathcal{R}(\mathcal{R}_m^{l-1}(\mathbb{A}, \mathcal{S}))$ for $l \geq 1$, initialized with $\mathcal{R}_m^0(\mathbb{A}, \mathcal{S}) \triangleq \mathcal{S}$.

5.1 Short-Horizon Switched MPC for $N_m^{\max} > \tau_{d,m}$

For the case of $N_m^{\max} > \tau_{d,m}$, one can directly select a minimum horizon $N_m^{\min} \geq N_m^{\max} - \tau_{d,m}$. The short-horizon switched MPC scheme is formulated as follows:

$$\mathbf{u}_k^* = \arg \min_{\mathbf{u}_k} J_m(x_k, \mathbf{u}_k) \quad (6a)$$

$$\text{s.t. } x_{i+1|k} = A_m x_{i|k} + B_m u_{i|k}, \quad x_{0|k} = x_k, \quad (6b)$$

$$x_{i|k} \in \mathbb{X}_m, \quad u_{i|k} \in \mathbb{U}_m, \quad \forall i \in \mathbb{Z}_{[0, N_m^m - 1]}, \quad (6c)$$

$$x_{i|k} \in \Theta(\lambda, s), \quad \forall i \in \mathbb{Z}_{[\delta_k, N_m^m]}, \quad (6d)$$

$$x_{N_m^m|k} \in \mathcal{R}_m^{\delta_k}(\Xi_\infty^\lambda, \mathcal{T}_\infty), \quad (6e)$$

where $\delta_k = \max\{\Delta \mathcal{N}_m^* - \mathcal{D}(k), 0\}$, $\Delta \mathcal{N}_m^* = \mathcal{N}_m^* - N_m^{\max}$, and $\mathcal{N}_k^m = \begin{cases} N_m^{\max}, & \mathcal{D}(k) \leq \Delta \mathcal{N}_m^*, \\ \max\{\mathcal{N}_m^* - \mathcal{D}(k), N_m^{\min}\}, & \mathcal{D}(k) > \Delta \mathcal{N}_m^*. \end{cases}$

Ideally, the state converges into the terminal set $\mathcal{T}_\infty$ within $\mathcal{N}_m^*$ steps. However, since the allowable maximum prediction horizon is strictly bounded by $N_m^{\max} < \mathcal{N}_m^*$, the controller cannot sense the terminal set directly at the switching instant. To bridge this temporal gap and ensure the state systematically reaches $\mathcal{T}_\infty$, intermediate predecessor sets are introduced. Once the mode duration becomes sufficiently long, this short horizon scheme smoothly reduces to the standard variable horizon MPC framework. For detailed schematic diagrams and step-by-step constraint evolution tables, one is referred to the extended version of this paper by Zhao et al. (2026).

Theorem 3 Consider the short-horizon switched MPC problem (6) for the switched system (1). If the horizon parameters satisfy $\mathcal{N}_m^* - N_m^{\min} \leq \tau_{d,m} < N_m^{\max}$, the dwell-time $\tau_{d,m}$ fulfills (5), and $x_0 \in \mathcal{F}_{\sigma(0)}$, where $\mathcal{F}_m$ is constructed by Algorithm 2 using the prediction horizon $\mathcal{N}_m^*$, then the switched system (1) is asymptotically stable.

The proof is omitted for brevity and is provided in the extended version of this paper by Zhao et al. (2026).

5.2 Short-Horizon Switched MPC for $N_m^{\max} \leq \tau_{d,m}$

Consider the case where $N_m^{\max} \leq \tau_{d,m}$. Under this condition, setting $\underline{\mathcal{N}}_m^* = N_m^{\min}$ is no longer feasible; thus, a modified MPC formulation is required. For any state constraint $x_{i|k} \in \mathcal{S}$ over $i \in \mathbb{Z}_{[a,b]}$, the constraint is void if $a > b$. Define $\mathcal{R}_m^i(\mathbb{X}_m, \Theta(\lambda, s)) = \mathcal{R}_m^i(\mathbb{X}_m, \Theta(\lambda, s))$.

The short-horizon MPC problem is formulated as follows:

$$\mathbf{u}_k^* = \arg \min_{\mathbf{u}_k} J_m(x_k, \mathbf{u}_k) \quad (7a)$$

$$\text{s.t. } x_{i+1|k} = A_m x_{i|k} + B_m u_{i|k}, \quad x_{0|k} = x_k, \quad (7b)$$

$$x_{i|k} \in \mathbb{X}_m, \quad u_{i|k} \in \mathbb{U}_m, \quad \forall i \in \mathbb{Z}_{[0, N_m^{\max} - 1]}, \quad (7c)$$

$$x_{i|k} \in \Theta(\lambda, l), \quad \forall i \in \mathbb{Z}_{[\varepsilon_k, \mathcal{N}_k^m - 1]}, \quad (7d)$$

$$x_{N_m^{\max}|k} \in \mathcal{R}_m^{\rho_k}(\mathbb{X}_m, \mathcal{R}_m^{\delta_k}(\Xi_\infty^\lambda, \mathcal{T}_\infty)), \quad (7e)$$

where $\varepsilon_k = \max\{\mathcal{N}_m^* - \underline{\mathcal{N}}_m^* - \mathcal{D}(k), 0\}$, $\rho_k = \max\{\mathcal{N}_m^* - \underline{\mathcal{N}}_m^* + N_m^{\max} - \mathcal{D}(k), 0\}$, $\delta_k = \begin{cases} \varepsilon_k & \rho_k = 0, \\ \mathcal{N}_m^* - \underline{\mathcal{N}}_m^* & \rho_k > 0, \end{cases}$ and the horizon length $\mathcal{N}_k^m$ is determined by

$$\mathcal{N}_k^m = \begin{cases} N_m^{\max}, & \mathcal{D}(k) \leq \Delta \mathcal{N}_m^*, \\ \max\{\mathcal{N}_m^* - \mathcal{D}(k), N_m^{\min}\}, & \mathcal{D}(k) > \Delta \mathcal{N}_m^*. \end{cases}$$

Note that, if $\delta(k, \mathcal{D}(k)) > \mathcal{N}_k^m$, the state constraint (7d) becomes inactive. For a detailed step-by-step illustration of the constraint evolution and the specific design of the terminal sets, one is referred to Table B.2 in the Appendix of Zhao et al. (2026).

Theorem 4 Consider the short-horizon switched MPC problem (7). Suppose that the dwell time $\tau_{d,m}$ satisfies (5), the horizon parameters satisfy $\mathcal{N}_m^* - \underline{\mathcal{N}}_m^* \leq \tau_{d,m}$, and the initial state $x_0 \in \mathcal{F}_{\sigma(0)}$. If the domain of attraction $\mathcal{F}_m$ is computed by Algorithm 2 using the maximum horizon $\mathcal{N}_m^*$, then the closed-loop system is asymptotically stable.

The proof parallels that of Theorem 3 and is omitted for brevity. Evidently, problem (7) relaxes the coupling between the dwell time and the prediction horizon bounds.

Remark 7 The proposed MPC formulations (3) typically yield a smaller domain of attraction compared to (Zhang et al. (2016)). To overcome this, the proposed short-horizon scheme strategically enlarges the initial dwell time $\hat{\tau}_{d,m}$ relative to the values of $\tau_{d,m}$ used in Algorithm 1. For fixed $N_m^{\min}$ and $N_m^{\max}$, the desired maximum horizon is defined as $\mathcal{N}_m^* = N_m^{\min} + \hat{\tau}_{d,m}$. Depending on the relationship between $N_m^{\max}$ and $\hat{\tau}_{d,m}$, the control sequence is computed via (6) or (7). The resulting domain of attraction coincides exactly with $\mathcal{F}_m$ generated by Algorithm 2 under $\mathcal{N}_m^*$. This confirms that the proposed method systematically expands the feasible region, validated by the subsequent simulations.

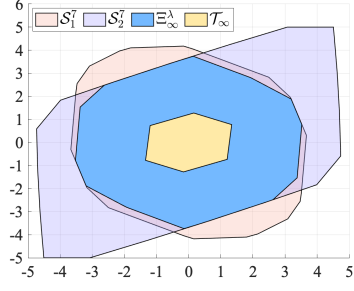
6 Simulations

Consider a switched system consisting of two subsystems

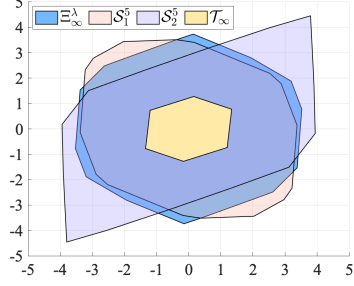
$$\text{with parameters } A_1 = \begin{bmatrix} 0.1397 & -0.9598 \\ 1.4822 & 0.3995 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix},$$

$$A_2 = \begin{bmatrix} 0.4557 & -1.0797 \\ 1.1777 & -0.4360 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The state and control constraints are given by $\mathbb{X}_1 = \mathbb{X}_2 = \{x \in \mathbb{R}^2 : \|x\|_\infty \leq 5\}$, $\mathbb{U}_1 = \{u \in \mathbb{R} : \|u\| \leq 1.5\}$ and $\mathbb{U}_2 = \{u \in \mathbb{R} : \|u\| \leq 1\}$. The state weighting matrices are $Q_1 = Q_2 = I$, and the input weighting matrices are $R_1 = R_2 = 1$. The terminal penalty matrices P_m and feedback gains K_m ($m \in \{1, 2\}$) are derived using the unconstrained LQR method. According to Lemma 2, the dwell-time constraints are $\tau_{d,1} = \tau_{d,2} = 2$. Set the horizons to $N_1^{\min} = N_2^{\min} = 5$, and $N_1^{\max} = N_2^{\max} = 7$.

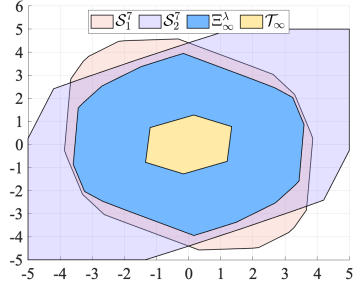


(a) Maximum prediction horizon

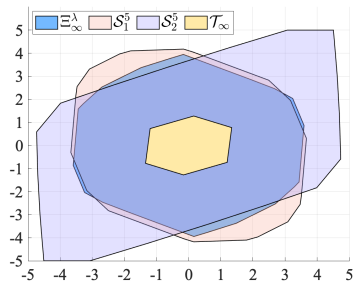


(b) Minimum prediction horizon

Fig. 3. The feasible regions with $N_1^{\max} = N_2^{\max} = 7$ and $N_1^{\min} = N_2^{\min} = 5$ for the standard MPC and the corresponding switching feasible set



(a) Maximum prediction horizon



(b) Minimum prediction horizon

Fig. 4. The feasible regions for the standard MPC with $N_1^{\max} = N_2^{\max} = 9$ and $N_1^{\min} = N_2^{\min} = 7$ and the corresponding switching feasible set

- *The relationship between the set Ξ_∞^λ and the set $\mathcal{S}_m^n$:* The set $\mathcal{S}_m^n = \mathcal{P}_m^n(\mathcal{T}_\infty)$ defines the n -step domain of attraction for the m -th subsystem under MPC constraints (3b), (3c), and (3e). As discussed in Remark 1, the switching feasible set Ξ_∞^λ is a subset of the maximal-

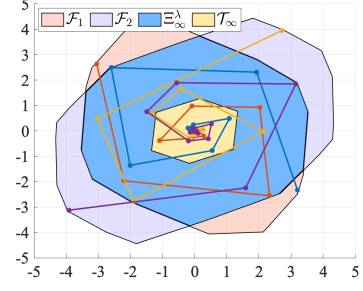
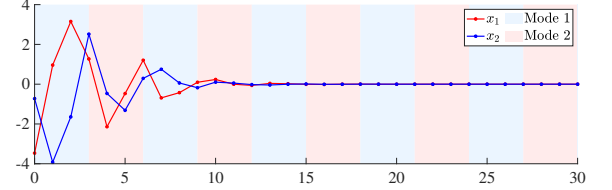
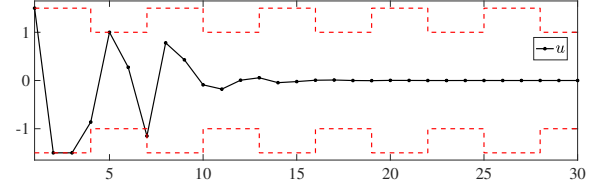


Fig. 5. The domains of attraction and the state trajectories from four different initial points



(a) State trajectories



(b) Control input

Fig. 6. State and input trajectories with the system initialized in subsystem 1

horizon intersection $\bigcap_{m \in \mathcal{M}} \mathcal{S}_m^{N_m^{\max}}$. Furthermore, there is no explicit inclusion relationship between Ξ_∞^λ and the minimal-horizon intersection $\bigcap_{m \in \mathcal{M}} \mathcal{S}_m^{N_m^{\min}}$.

These geometric relationships are visually demonstrated by Figs. 3 and 4. Fig. 3 plots Ξ_∞^λ together with the sets $\mathcal{S}_m^n$ for all $m \in \mathcal{M}$ and $n \in \{N_m^{\min}, N_m^{\max}\}$ (set to 5 and 7, respectively). The set Ξ_∞^λ is strictly contained in the maximal-horizon intersection $\mathcal{S}_1^7 \cap \mathcal{S}_2^7$, whereas the minimal-horizon inclusion $\mathcal{S}_1^5 \cap \mathcal{S}_2^5 \subseteq \Xi_\infty^\lambda$ is not guaranteed. Fig. 4 evaluates the sets with extended prediction horizons of $N_m^{\max} = 9$ and $N_m^{\min} = 7$. As shown, neither $\mathcal{S}_1^7 \cap \mathcal{S}_2^7 \subseteq \Xi_\infty^\lambda$ nor $\Xi_\infty^\lambda \subseteq \mathcal{S}_1^7 \cap \mathcal{S}_2^7$ holds.

- *The effectiveness of the variable-horizon switched MPC:* Fig. 5 depicts the domains of attraction derived from *Algorithm 2*, superimposed with state trajectories starting from four distinct initial conditions. Under the mode-dependent dwell-time constraint, every initial state in the domain of attraction is feasible for the proposed switched MPC problem (3). Quantitatively, the switching feasible set has a volume of 38.5133 and is strictly contained within $\bigcap_{m \in \mathcal{M}} \mathcal{F}_m$, which has a volume of 38.7460. Figs. 6 and 7 illustrate the state trajectories, control inputs, and the corresponding switching signals when the system is initialized in subsystem 1 and subsystem 2, respectively. As shown, all state trajectories successfully converge to the origin within 20 time steps.

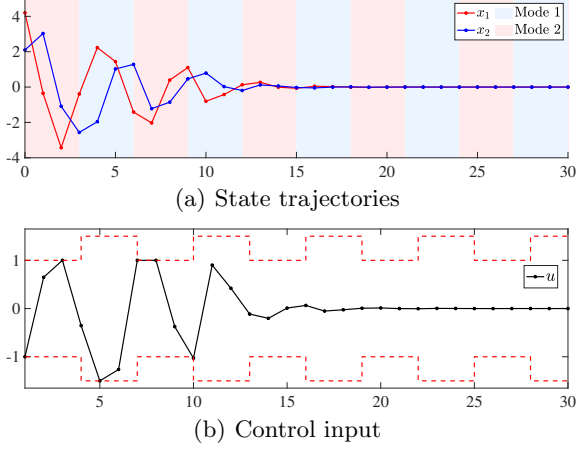


Fig. 7. State and input trajectories with the system initialized in subsystem 2

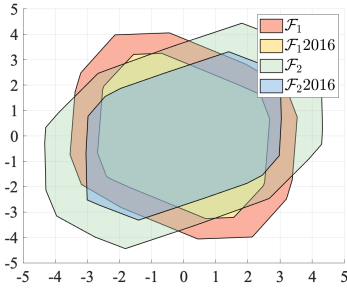


Fig. 8. The domains of attraction using the MPC proposed in this paper and the method in Ong et al. (2016) for switched system in this paper

- *Comparison with (Ong et al. (2016))*: Fig. 8 illustrates the domains of attraction obtained using the proposed method and the method in Ong et al. (2016). $\mathcal{F}_m$ and $\mathcal{F}_m^{2016}$ represent the domains of attraction for the proposed method and the scheme in Ong et al. (2016), respectively. The key observations are as follows: 1) $\mathcal{F}_m^{2016}$ is a strict subset of $\mathcal{F}_m$, meaning that every state admissible under the method in Ong et al. (2016) is also feasible for the proposed switched MPC (whereas the converse is false), which naturally results in a strictly larger volume for $\mathcal{F}_m$; 2) The proposed MPC scheme is computationally more efficient as it bypasses the max-min optimization required in Ong et al. (2016).
- *Comparison with (Zhang et al. (2016))*: The dwell-time constraints adopted here exhibit greater flexibility than those in Zhang et al. (2016). Table 2 compares the dwell-time constraints required by the method in

Table 2
Comparison of dwell-time constraints using the method in Zhang et al. (2016) and the proposed scheme

DT	Zhang et al. (2016)								Proposed
Stage	1	2	3	4	5	6	7	≥ 8	≥ 1
$\tau_{d,1}$	5	5	5	4	4	3	4	2	2
$\tau_{d,2}$	4	4	4	2	2	2	2	2	2

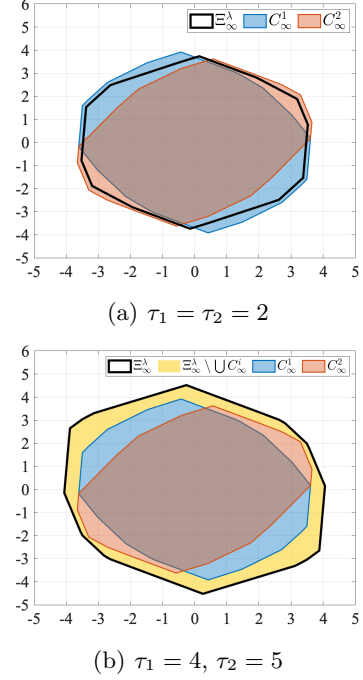


Fig. 9. The switching feasible set Ξ_∞^λ of Algorithm 1 and the switch-RCI sets C_∞^i of Algorithm 1 in Danielson et al. (2019)

Zhang et al. (2016) and the proposed approach. The constraints in Zhang et al. (2016) are derived through complex set operations. These operations increase the offline computational burden, while the requirement to ensure feasibility across all admissible mode transitions may lead to conservative dwell-time bounds. Moreover, their calculation procedure is computationally demanding, as the dwell-time bounds are strongly coupled with the number of switches. In contrast, the proposed MPC scheme yields less restrictive dwell-time constraints. Furthermore, when the system matrices from (Zhang et al. (2016)) are evaluated within our framework, the required dwell time is significantly reduced to $\tau_{d,1} = 2$ and $\tau_{d,2} = 4$. Notably, these shortened durations would be completely infeasible under the stringent conditions established in Zhang et al. (2016), clearly demonstrating the expanded operational flexibility of our approach.

- *Comparison with (Danielson et al. (2019))*: To visually assess the geometric properties of the feasible regions, Fig. 9 compares our switching feasible set Ξ_∞^λ with the exact switch-RCI sets C_∞^i computed by the method in Danielson et al. (2019). As discussed in Remark 2, these two methods impose different set conditions, and neither definition implies a general inclusion relationship. The switch-RCI construction enforces one-step control invariance, whereas the proposed method allows the state to temporarily leave Ξ_∞^λ provided that the physical constraints remain satisfied and the prescribed multi-step transition is completed. The effect of dwell time can be observed by comparing the two parameter settings. As seen in Fig. 9(a), under short dwell-time constraints ($\tau_1 = \tau_2 = 2$), the feasible regions of both methods are

comparable. However, when the dwell time increases to $\tau_1 = 4$ and $\tau_2 = 5$ in Fig. 9(b), the proposed switching feasible set expands, whereas the switch-RCI sets remain constrained by their one-step invariance conditions. Conversely, the proposed algorithm uses the additional dwell-time steps to retain additional states in Ξ_∞^λ , as evidenced in this example.

- *Comparative evaluation and computational complexity analysis:* To evaluate the conservatism and computational efficiency of the proposed scheme, comprehensive comparisons were conducted against existing approaches (Danielson et al. (2019); Ong et al. (2016); Zhang et al. (2016)). Additional system parameters are provided in the Appendix. To ensure fairness, the prediction horizon for the reference methods was uniformly set to $N = 5$. The offline computation time was averaged over 10 independent runs. For online MPC optimization, initial states were randomly sampled from the intersection of the initial feasible sets of all evaluated methods. A unified switching signal, derived from the most conservative constraints in Zhang et al. (2016), was applied, and the average computation time for a single optimization step was recorded over every 20 steps.

Regarding computational efficiency, the proposed offline phase requires slightly more time than that of Ong et al. (2016), owing to the explicit calculation of the switching feasible set. However, it exhibits a significant speed advantage over the methods in Zhang et al. (2016) and Danielson et al. (2019). This advantage becomes particularly prominent as the number of subsystems (M) or state dimensions (n_x) increases; notably, for $n_x = 3$, the method in Danielson et al. (2019) failed to converge within 300 iterations (exceeding 500 s). During the online phase, our computation time is marginally higher than the baseline in Zhang et al. (2016). This slight overhead is fundamentally attributed to the longer prediction horizon inherently supported by our method. If the horizon in Zhang et al. (2016) is extended to match (e.g., $N = 7$ for $M = 2, n_x = 2$), their online calculation time rises to 0.1709 s, demonstrating that our online efficiency remains highly competitive. Overall, the proposed method achieves a favorable offline-online computational trade-off in the tested configurations, reducing offline computation time relative to set-based methods while maintaining comparable online efficiency.

- *The application of short-horizon switched MPC:* The domains of attraction reported in Zhang et al. (2016), configured with $N_1^{\max} = N_2^{\max} = 7$, are larger than those proposed standard variable-horizon MPC (3). This conservative limitation is effectively overcome by the proposed *short-horizon* switched MPC (7). In Fig. 10, $\hat{\mathcal{F}}_m$ and $\tilde{\mathcal{F}}_m^{N_m^*}$ represent the domains of attraction obtained using the method in Zhang et al. (2016) and the *short-horizon* switched MPC, respectively. The remaining parameters are specified as $N_1^{\min} = N_2^{\min} = 5$, $N_1^{\max} = N_2^{\max} = 7$, $N_1^* = 11$, and $N_2^* = 10$. For the first switching instant, the dwell-time constraints $\hat{\tau}_{d,1} = 6$ and

Table 3

Quantitative comparison of computation times

Scenario / Method		Offline(s)	Online(s)
$M = 2;$ $n_x = 2$	Proposed	7.185	0.1697
	Ong et al. (2016)	1.656	0.3730
	Zhang et al. (2016)	29.051	0.1669
	Danielson et al. (2019)	37.015	0.1662
$M = 4;$ $n_x = 2$	Proposed	8.133	0.1832
	Ong et al. (2016)	3.068	1.6579
	Zhang et al. (2016)	34.434	0.1796
	Danielson et al. (2019)	268.422	0.1789
$M = 2;$ $n_x = 3$	Proposed	250.150	0.1791
	Ong et al. (2016)	80.614	0.3526
	Zhang et al. (2016)	496.464	0.1845
	Danielson et al. (2019)	—	—

$\hat{\tau}_{d,2} = 5$ align with those in Zhang et al. (2016). Notably, the *short-horizon* switched MPC delivers strictly larger domains of attraction than those reported in Zhang et al. (2016). This demonstrates the enhanced capability of the proposed approach to enlarge the feasible region. Finally, Fig. 11 plots the state trajectories under the *short-horizon* switched MPC (6), confirming the closed-loop asymptotic stability of the system.

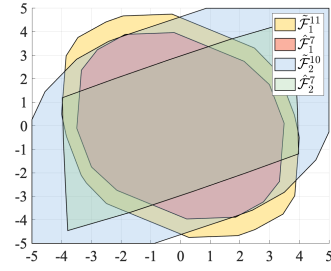


Fig. 10. The domains of attraction using $N_1^{\min} = N_2^{\min} = 5$, $N_1^* = 11$ and $N_2^* = 10$.

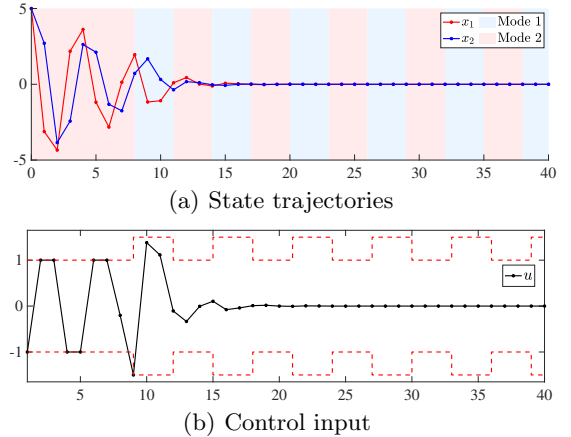


Fig. 11. State and input trajectories under the short-horizon switched MPC

7 Conclusion

This paper presents a *variable-horizon* switched MPC scheme for switched systems. An algorithm is devel-

oped to compute the switching feasible set. By leveraging this set, persistent feasibility is guaranteed, provided that the conditions regarding the prediction horizon and dwell-time constraints are satisfied. The dwell-time conditions are derived from the corresponding unconstrained closed-loop subsystems, which also ensure the stability of the switched systems under the proposed MPC. Furthermore, two *short-horizon* switched MPC schemes are proposed. They can relax the constraint that dwell-time is less than the maximum prediction horizon, thereby expanding the domain of attraction. Simulations with comparisons illustrate the performance of the proposed MPC scheme for switched systems.

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A System Parameters for Simulations

A.1 2D Switched System with four modes

To evaluate the proposed algorithm on a system with more modes, two additional subsystems (Modes 3 and 4) are appended to the 2D system, given by

$$A_3 = \begin{bmatrix} -0.0724 & 0.9009 \\ 0.6191 & 0.5519 \end{bmatrix}, \quad B_3 = \begin{bmatrix} 0.2 \\ -0.5 \end{bmatrix},$$

$$A_4 = \begin{bmatrix} 0.8678 & 0.3657 \\ 0.8176 & -0.3591 \end{bmatrix}, \quad B_4 = \begin{bmatrix} -0.3 \\ 0.5 \end{bmatrix}.$$

The state and control constraints are uniform for both modes, given by $\mathbb{X}_3 = \mathbb{X}_4 = \{x \in \mathbb{R}^2 : \|x\|_\infty \leq 5\}$ and $\mathbb{U}_3 = \mathbb{U}_4 = \{u \in \mathbb{R} : |u| \leq 1\}$. The weighting matrices are set to $Q_3 = Q_4 = I$ and $R_3 = R_4 = 4$. The terminal penalty matrices P_m and feedback gains K_m ($m \in \{3, 4\}$) are derived via unconstrained LQR. The required dwell time is $\tau_{d,3} = \tau_{d,4} = 2$. The variable prediction horizons are bounded by $N_3^{\min} = N_4^{\min} = 5$ and $N_3^{\max} = N_4^{\max} = 7$.

A.2 3D Switched System

For higher-dimensional test, a 3D switched system comprising two subsystems is considered. The system dynamics are

$$A_1 = \begin{bmatrix} -0.8125 & 0.7000 & -0.9375 \\ -0.8125 & -0.2375 & 0.1250 \\ -1.2375 & 0.1250 & 0.9500 \end{bmatrix}, \quad B_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix},$$

$$A_2 = \begin{bmatrix} -0.7750 & 0.9375 & 0.0125 \\ 0.6125 & 1.1750 & -0.4375 \\ -0.6875 & 0.8000 & -0.4750 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 1 \\ 1 \\ -0.5 \end{bmatrix}.$$

The state constraints have the forms of $\mathbb{X}_1 = \{x \in \mathbb{R}^3 : \|x\|_\infty \leq 10\}$ and $\mathbb{X}_2 = \{x \in \mathbb{R}^3 : \|x\|_\infty \leq 9\}$. The control inputs are bounded in $\mathbb{U}_1 = \{u \in \mathbb{R} : \|u\| \leq 1.5\}$ and $\mathbb{U}_2 = \{u \in \mathbb{R} : \|u\| \leq 2\}$. The LQR weighting matrices are specified as $Q_1 = 10I$, $Q_2 = 5I$, and $R_1 = R_2 = 1$, which similarly yield the corresponding P_m and K_m . The horizon bounds are chosen as $N_1^{\min} = N_2^{\min} = 5$ and $N_1^{\max} = N_2^{\max} = 7$.

B Short-Horizon Switched MPC for $N_m^{\max} > \tau_{d,m}$

A schematic diagram for the short-horizon switched MPC problem (6) is illustrated by Fig. A.1, with Table B.1 detailing the prediction horizon and state constraints at each step. Ideally, the state should enter the terminal set $\mathcal{T}_\infty$ within N_m^* steps. However, since the allowable maximum prediction horizon satisfies $N_m^{\max} < N_m^*$, only a restricted prediction window (the red dashed box at k_s) is available at the switching instant. To bridge this temporal gap, intermediate transitional sets (the purple regions in Fig. A.1) are introduced as multi-step predecessor sets of $\mathcal{T}_\infty$.

The key distinction from the standard variable-horizon MPC (Fig. 2) lies precisely in this transitional phase. Once the dwell time after the switch becomes sufficiently long (e.g., at $k_s + 2$, represented by the green dashed box), the short-horizon scheme smoothly reduces to the standard variable-horizon MPC. A direct comparison between Tables 1 and B.1 reveals that the terminal

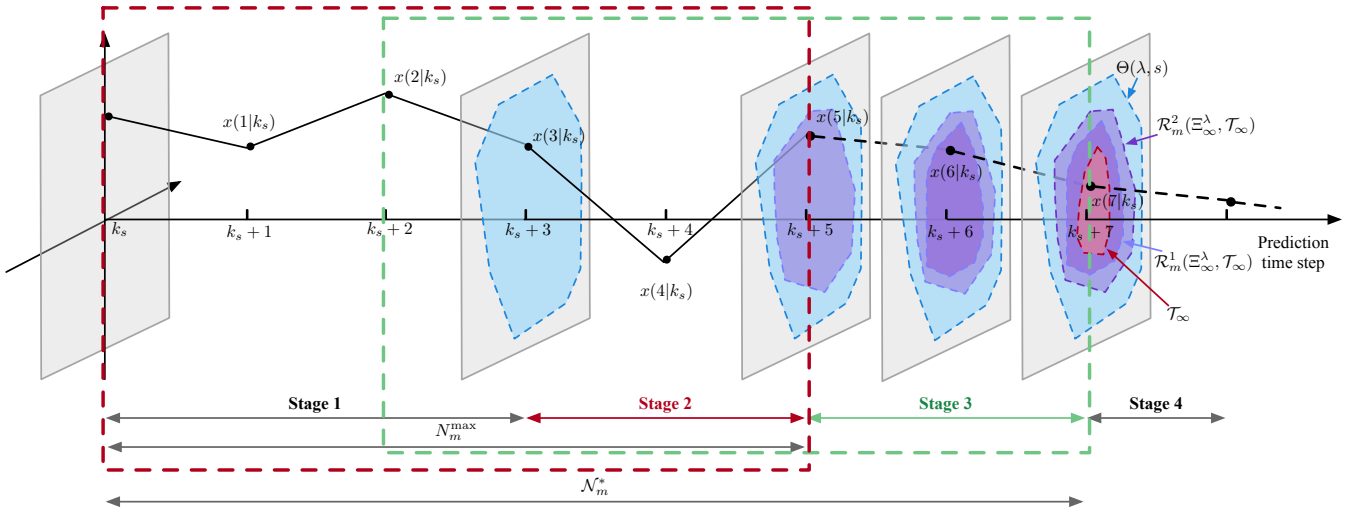


Fig. A.1. Multi-mode prediction paradigm of the short-horizon switched MPC with $N_m^* = 7$, $N_m^{\max} = 5$, and $N_m^{\min} = 2$. The red dashed box shows the prediction at the switching instant k_s , while the green dashed box (at $k_s + 2$) coincides with the standard variable-horizon MPC in Fig. 2. At k_s , the control sequence steers the predicted state into $\Theta(\lambda, s)$ from step $N_m^{\max} - N_m^{\min}$ onward, and forces the $N_m^{\max}$ -step-ahead state into the predecessor set $\mathcal{R}_m^{\rho k}(\Xi_\infty^\lambda, \mathcal{T}_\infty)$, ensuring arrival at $\mathcal{T}_\infty$ within N_m^* steps. The purple region realizes a transitional phase: although the prediction horizon is shorter than N_m^* , the nested predecessor structure guarantees that the terminal requirement is met once the mode duration is sufficiently long. After entering $\mathcal{T}_\infty$, the predicted trajectory follows the local LQR law $u(k) = K_{\sigma(k)}x(k)$.

Table B.1

Prediction horizon and state constraints at each instant and predicted step with $\mathcal{N}_m^* = 7$, $N_m^{\max} = 5$, $N_m^{\min} = 2$ (see (6))

	k_s	$k_s + 1$	$k_s + 2$	$k_s + 3$	$k_s + 4$	$[k_s + 5, k_{s+1}]$
Horizon	5	5	5	4	3	2
Step 1	$\mathbb{X}_m$	$\mathbb{X}_m$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$
Step 2	$\mathbb{X}_m$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\mathcal{T}_\infty$
Step 3	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\mathcal{T}_\infty$	N/A
Step 4	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\mathcal{T}_\infty$	N/A	N/A
Step 5	$\mathcal{R}_m^2(\Xi_\infty^\lambda, \mathcal{T}_\infty)$	$\mathcal{R}_m^1(\Xi_\infty^\lambda, \mathcal{T}_\infty)$	$\mathcal{T}_\infty$	N/A	N/A	N/A

Table B.2

Prediction horizon and state constraints at each instant and predicted step, with $\mathcal{N}_m^* = 8$, $\underline{\mathcal{N}}_m^* = 3$, $N_m^{\max} = 4$, and $N_m^{\min} = 2$ (see (7))

	k_s	$k_s + 1$	$k_s + 2$	$k_s + 3$	$k_s + 4$	$k_s + 5$	$[k_s + 6, k_{s+1}]$
Horizon	4	4	4	4	4	3	2
Step 1	$\mathbb{X}_m$	$\mathbb{X}_m$	$\mathbb{X}_m$	$\mathbb{X}_m$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$
Step 2	$\mathbb{X}_m$	$\mathbb{X}_m$	$\mathbb{X}_m$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\mathcal{T}_\infty$
Step 3	$\mathbb{X}_m$	$\mathbb{X}_m$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\Theta(\lambda, s)$	$\mathcal{T}_\infty$	N/A
Step 4	$\mathcal{R}_m^1(\mathbb{X}_m, \mathcal{R}_m^3(\Xi_\infty^\lambda, \mathcal{T}_\infty))$	$\mathcal{R}_m^3(\Xi_\infty^\lambda, \mathcal{T}_\infty)$	$\mathcal{R}_m^2(\Xi_\infty^\lambda, \mathcal{T}_\infty)$	$\mathcal{R}_m^1(\Xi_\infty^\lambda, \mathcal{T}_\infty)$	$\mathcal{T}_\infty$	N/A	N/A

constraint in the short-horizon formulation is no longer $\mathcal{T}_\infty$ itself, but a recursively constructed predecessor set $\mathcal{R}_m^{\rho_k}(\Xi_\infty^\lambda, \mathcal{T}_\infty)$, which guarantees that the state reaches $\mathcal{T}_\infty$ exactly at step $\mathcal{N}_m^*$.

PROOF. Persistent Feasibility: During the interval $[k_0, k_0 + \mathcal{N}_m - N_m^{\max})$, one has $\mathcal{D}(k) < \mathcal{N}_m - N_m^{\max}$ and (6) reduces to a fixed-horizon MPC with horizon length $N_m^{\max}$ where $m = \sigma(k_0)$. By the definition, $\mathcal{R}_m^{\delta_k}(\Xi_\infty^\lambda, \mathcal{T}_\infty)$ ensures that an admissible control drives the predicted state into $\mathcal{R}_m^{\delta_{k+1}}(\Xi_\infty^\lambda, \mathcal{T}_\infty) = \mathcal{R}_m^{\delta_k-1}(\Xi_\infty^\lambda, \mathcal{T}_\infty)$. Consequently, feasibility at instant k guarantees the feasibility of the problem at instant $k + 1$. Iterating this logic over the initial interval ensures recursive feasibility, while feasibility for subsequent steps follows directly from Theorem 1.

Stability: The feasibility of (6) requires an input sequence $u_{i|k} \in \mathbb{U}_m$ ($i \in \mathbb{Z}_{[0, N_m^{\max}-1]}$) such that 1) $x_{i|k} \in \mathbb{X}_m$ ($i \in \mathbb{Z}_{[1, \Delta \mathcal{N}_m^*-1]}$); 2) $x_{N_m^{\max}|k} \in \mathcal{R}_m^{\delta_k}(\Xi_\infty^\lambda, \mathcal{T}_\infty)$. By the recursive nature of the predecessor operator and the construction of Algorithm 2, the set of all initial states $x_{0|k}$ satisfying these exact multi-step constraints is precisely the domain of attraction $\mathcal{F}_m$. Thus, problem (6) is mathematically equivalent to the standard switched MPC problem (3) configured with maximum horizon $\mathcal{N}_m^*$ and minimum horizon $N_m^{\min}$. By Theorem 2, asymptotic stability is guaranteed. $\square$

C Short-Horizon Switched MPC for $N_m^{\max} \leq \tau_{d,m}$

Table C.1 compares the required minimum dwell-time constraints obtained from the method in Zhang et al. (2016) and the proposed short-horizon switched MPC scheme for a switched linear system with four subsystems. A further comparison with the two-subsystem case

Table C.1

Comparison of dwell-time constraints using the method in Zhang et al. (2016) and the proposed scheme

DT	Zhang et al. (2016)							Proposed
	1	2	3	4	5	6	≥ 7	
Stage								
$\tau_{d,1}$	7	3	3	4	4	4	2	2
$\tau_{d,2}$	7	6	6	4	2	4	2	2
$\tau_{d,3}$	6	6	5	4	4	4	4	2
$\tau_{d,4}$	2	2	2	2	2	2	2	2

in Table 2 reveals a notable limitation of the method of Zhang et al. (2016). Even when the first two subsystem matrices (A_1, B_1) and (A_2, B_2) remain identical, the dwell-time constraints computed by Zhang et al. (2016) change significantly simply due to the increase in the total number of subsystems. More critically, these variations exhibit no clear monotonicity or physical interpretability. For example, in the two-subsystem scenario, $\tau_{d,1}$ eventually converges to 2 after stage 8, whereas in the four-subsystem case, $\tau_{d,1}$ fluctuates between 2 and 7 across different stages despite the unchanged subsystem dynamics. This phenomenon indicates that the conservatism of Zhang et al. (2016) originates not from the intrinsic properties of individual subsystems, but from the combinatorial complexity of multi-step reachable-set operations across all modes.

In sharp contrast, the proposed scheme yields a consistent dwell-time requirement $\tau_{d,i} = 2$ for all subsystems, regardless of the total number of modes or the prediction stage. This stage-invariant and subsystem-independent result demonstrates that the proposed method successfully decouples the dwell-time condition from the combinatorial explosion inherent in switched systems, thereby offering significantly improved scalability and switching flexibility.